DL

- Binary Cross entropy (BCE) Loss function

$$L = -\left[y\log(\hat{y}) + (1-y)\log(1-\hat{y})\right]$$

y = True label (0 or 1)

$\hat{y}$ = Predicted probability

$$\frac{\partial L}{\partial \hat{y}} = -\left[y\frac{1}{\hat{y}} + (1-y)\frac{-1}{1-\hat{y}}\right]$$

$$= -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

$$= \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})}$$

$$= \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})}$$

$$= \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$